AFFINE FIBRES, BASE CHANGE AND THE UNIVERSAL MAP FOR A RELATIVE SPEC

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Allowed standard prerequisites (editorial).

- Glueing description. lemma-glue-relative-spec (same chapter, judged non-self-contained this batch, not carded). Affine-local description of the relative spectrum, giving part (1).
- Base change of F. lemma-spec-base-change (carded this batch). Functor base change, giving part (2).
- Affine case. lemma-spec-affine (same chapter, judged non-self-contained this batch, not carded). Part (3) over an affine base: the universal map is an isomorphism.

Source reference. Original lemma 01LP: sources/stacks-constructions/pinned-original-source.tex, source lines 599–599. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 01LT: sources/stacks-constructions/pinned-original-source.tex, source lines 709–709. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 01LS: sources/stacks-constructions/pinned-original-source.tex, source lines 685–685. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let S be a scheme. Let $\mathcal{A}$ be a quasi-coherent sheaf of $\mathcal{O}_S$-algebras. Let $\pi : \underline{\mathrm{Spec}}_S(\mathcal{A}) \to S$ be the relative spectrum of $\mathcal{A}$ over S.*

(1) *For every affine open $U \subset S$ the inverse image $\pi^{-1}(U)$ is affine.*
(2) *For every morphism $g : S' \to S$ we have $S' \times_S \underline{\mathrm{Spec}}_S(\mathcal{A}) = \underline{\mathrm{Spec}}_{S'}(g^*\mathcal{A})$.*
(3) *The universal map*

$$\mathcal{A} \longrightarrow \pi_*\mathcal{O}_{\underline{\mathrm{Spec}}_S(\mathcal{A})}$$

is an isomorphism of $\mathcal{O}_S$-algebras.

Proof. Part (1) comes from the description of the relative spectrum by glueing, see Lemma 01LP. Part (2) follows immediately from Lemma 01LS. Part (3) follows

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because it is local on S and it is clear in case S is affine by Lemma 01LT for example. $\square$